Team Bernie
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# Team Bernie Report

For this segment of the project the first step that my group did was clean the data. Looking at the document there were hundreds of thousands lines of data spanning over 50 years and we all brainstormed that it would be too tedious even to do 10 years. Instead we came to the decision that the years between 2010-2015 would be best for the cleanest and reliable and relevant data. If we did a 5 year span from 2015-2020 it would not be as credible due to the irregular data during the pandemic years so we decided to take data from earlier years for cleaner and more reliable results. My group was able to break down the numbers using python through Google Collab and collect the variables needed for the code. The variables that we needed to collect are the number of vehicles that are crossing to each border throughout the 5 year span which has thousands of vehicles that are crossing throughout the United States and the borders between Mexico and Canada. Additionally, even though data closer to 2025 would be more accurate the years in which the pandemic affected businesses would not. Between irregular shipping routes, workforce shortages, and borders that would close throughout the 2-3 year span would not give accurate data of a post pandemic world. We decided it would be best to get earlier data risk relevance for more accuracy in activity. We believed that it would be the best data to predict the outcome of the number of vehicles that are entering through each border.

When it came to running the data we ran into issues of it not being as precise as we desired it to be. The data was accurate and we even used AI such as Chat GBT to assist and back track possible errors we might have had that were unknown. I believe that this could have been the case since there were thousands of lines of data that was put into code and that possibly the site couldn't correctly depict that data as smoothly as if there was less data to interpret through. I believe this would have been the case with whichever years we would have picked since there was a lot of data and sometimes the code will not depict the most accurate of predictive models for the future. The way that any logic connects with the python is that it can create a model that will predict how many vehicles are crossing the border at a given time and specific day throughout different quarters of the year. These models are able to show not only one border but all borders that are between the United States. This type of predictive data is crucial for companies that ship international products since it can give them somewhat real data to predict possible shipping delays or how to avoid certain events from happening. This is why it is evenly important to find clean data to create a model about during a time in which shipping is normal if there aren't too many irregular events that would throw off data compared to present time.